NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Contents

Abstract . . . 3

1. One spatial seed, offset history and a generated geometry . . . 4

2. The smooth zero interface and its inherited source . . . 6

3. One source, three connected crossing numbers . . . 7

4. Pre-registered test: does the flow select today's vacuum fraction? . . . 8

5. What continues across zero: pressure and the late tail . . . 9

6. Finite amplitude: admissibility and the child criterion . . . 9

7. Selection and the charge-resolved question . . . 11

8. Outward status of the single-horizon control vs. the preserved atlas . . . 12

9. The Non-Local Multiverse to Local Observer Atlas (Axioms A0--A4) . . . 12

Appendix A. Technical atlas and preserved mathematical record . . . 19

Abstract

Nested-Space Cosmology (NSC) investigates whether one inherited geometric law can connect nested regions, their exchange of energy across event horizons, and the quantities measured by an interior observer. The foundation rests on five openly declared geometric axioms (**A0--A4**): (**A0**) geometry follows the path of least resistance by minimizing the 1D Cauchy--Schwarz generator stretch cost plus enclosing compression work, $E = Q_{stretch} + PV$, without fluid incompressibility or borrowed surface tension; (**A1**) a 1D spatial seed plus an inherited compression offset (time) generates a closed 2D circle and 3D sphere; (**A2**) seen edge-on, the closed circle $z(\tau)=R_0 e^{i\omega\tau}$ is the harmonic pendulum $R(\tau)=R_0 cos(\omega\tau)$, which crosses `0` smoothly into the negative direction $R<0$ (`-`), flipping the 3D orientation Jacobian $det\ F=(R/R_0)^3$ from `+` to `-` and turning inward implosion ($\dot{R}<0$) into outward child expansion ($\dot{r}_-=-\dot{R}>0$) without the artificial velocity cusp $\Delta\dot{r}=+2R_0\omega$ created when standard physics restricts radius to an unsigned size $r=|R|\geq 0$; (**A3**) each nested room carries a moving wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its Planck cutoff (λ_{min}) and Absolute-Zero horizon (λ_{max}); and (**A4**) every room sits between an outer birth `0` valve and inner black-hole `0` valves on a closed vortex--diffusion ring (`0 <-> o`).

Minimizing the unified 5-variable single-cost functional across `+`, `0`, and `-` yields a unique closed-form global minimum (Section 9) and six parameter-free geometric and dynamical theorems: (1) eliminating both internal variables (θ^*,χ) from the five-variable budget shares (D,M,V) with $B=M-D/3$ yields the exact zero-parameter algebraic curve $9DVB=4(V+B/2)^3$, inverse recovery $\theta^*=1/(1+2V/B)$ and $e^{2\chi}=(2V+B)/(B)\sqrt{((2V+B)/(3DV))}$, and the deep-nest invariant $lim_{\chi\to\infty} B/V^2=16/27$; (2) without fluid assumptions, the infalling `+` parcel shrinks in volume while spaghettifying into a 1D filament above $\chi_{spaghetti}=(1)/(2)ln\ 2\approx 0.346574$ ($\theta^*=1/2$), while three independent geometric mechanisms ($d\lambda/\lambda$ scale-invariant measure, active-tube concentration $w_{tube}=1$, and zero-mode topological phase $\Delta\theta/\pi=1/2$) bypass the e^{-3K} volumetric dilution wall with $O(1)$ survival, and 3D standing-wave closure selects the `(1,1,1)` triad ground state ($E_{1,1,1}=3\pi^2/R^2$, *2:1* split against the S^2 angular floor) with a finite wave-compression hard core $R_{core}=\sqrt{2\pi/\kappa}$; (3) for any incompressible irrotational potential flow around a spherical throat of radius $R(\tau)$ with volume flux $Q=4\pi R^2\dot{R}$, the exterior kinetic energy $T_{ext}=2\pi\rho_0 R^3\dot{R}^2$ and surrounding-flow Virial work $W_{flow}=\int_{r>R} x\cdot(-\nabla p)\, d^3x$ obey the exact parameter-free *3:1* partition $T_{ext}:W_{flow}=3:1$; (4) coupling a finite parent reservoir $E_P(\tau)+E_-(\tau)=E_{tot}$ produces the closed *3:1* relaxation law $R(\tau)/R_\infty=tanh(\sqrt{2E_0}\ \tilde{\tau})$ with zero-radius continuum opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$, a hybrid last-echo horizon cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot *25.63%*), and self-consistent parent curvature backreaction (*8.98%* asymptotic shift); (5) integrating the outward causal stress balance over the entire exterior $r\in[R,\infty)$ of the exact harmonic two-center family gives the parameter-free algebraic neck-splitting onset cubic $p^3-3p-3=0$ ($p_{crit}=1+2^{1/3}+2^{-1/3}\approx 2.52988065$, $(c/R)_{onset}=0.56786589<(c/R)_{pinch}=2^{-1/3}\approx 0.79370053$), finite-time pinch crossing $\Delta\tau_{pinch}=0.506704$, positive central observer pulse $max\ \varphi_C=+0.163689$, and exact analytical pole-sum tail $\varphi_C(T)=-0.14192836/T'+0.01750496/(T')^2+O((T')^{-3})$; and (6) harmonicity ($\Delta\Phi=0$) forces the directional enclosed-volume derivative trace $\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$ to vanish at every stage, driving a 3-axis orthogonal post-pinch cascade $e_z\to e_x\to e_y$ ($1\to 2\to 4\to 8$ lobes, exact harmonic ratios *-1/2* and *-2*) coupled by the exact multipole identity $\partial_{q_2} g_{l,0}|_{q_2=0}=-(l-1)/(2)\partial_{q_1} g_{l,0}$ and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix ($M_{12}=-0.0054817211\neq 0$, $det\ M_{full}=0.1400308435>0$, initial transverse conjugate momentum $p_2(0)=+0.05157014>0$). Secondary single-valve epoch and ladder ansätze ($\Omega_{vis}(\chi_*)=e^{-3}\approx 4.9787\%$ giving *71.1441% / 23.8772% / 4.9787%*, and $K=7$ giving $\eta_7=5.955329\times 10^{-10}$) are strictly quarantined in Section 9.7 from the parameter-free theorems above.

Sections 2--8 preserve the conditional two-scalar Einstein control apparatus and the pre-registered single-horizon flow test NSC-P006 (*88.4735067504%* scalar patch-energy crossing, normalized area growth $1.085638919782\ \varepsilon^2$, short-leg transfer $2.219782587242\ \varepsilon^2$, and one-mode neck-trapping threshold $\varepsilon_{crit}=0.8339066$), while Appendix A summarizes the nine preserved technical engines and links to the archived `v0.26.0` foundation record.

1. One spatial seed, offset history and a generated geometry

Douglas's proposal is that contrast and exchange organize approximately spherical regions, and that the construction can repeat within them. Time names the offset operation in this generating picture. A physical duration is a readout along the resulting history. The generated space has three spatial directions; a child is another region, not a fourth primitive spatial direction.

The circle and sphere separate an inside from an outside. Isotropic positive surface energy at fixed volume selects a sphere by the isoperimetric inequality. Whether a particular gravitational region is selected by that variational problem must follow from its action. The present smooth horizon has no material tension. A material-shell branch is therefore retained separately in the technical atlas.

The explicit two-stage offset generator is

$$\theta_{n+1} = \theta_n + \alpha$$
$$\phi_{n+1} = \phi_n + \theta_n$$

with phases taken modulo one and zero initial phases ($\theta_0=\varphi_0=0$). Hence $\theta_n=n\alpha$ and $\varphi_n=(1)/(2)n(n-1)\alpha$ modulo one. One accumulator feeds the second; no independent second phase is supplied. For irrational α, Weyl's criterion proves that the joint phase history is dense on the two-phase torus T^2. This uses the classical polynomial equidistribution framework, attributed in the generator proof.

Setting $\Theta_n=2\pi\theta_n$ and $\Phi_n=2\pi\varphi_n$, the generated unit directions and their inner-product kernel are

$$v_n = (\sin\Theta_n\cos\Phi_n, \sin\Theta_n\sin\Phi_n, \cos\Theta_n)$$
$$C_{nm} = v_n \cdot v_m.$$

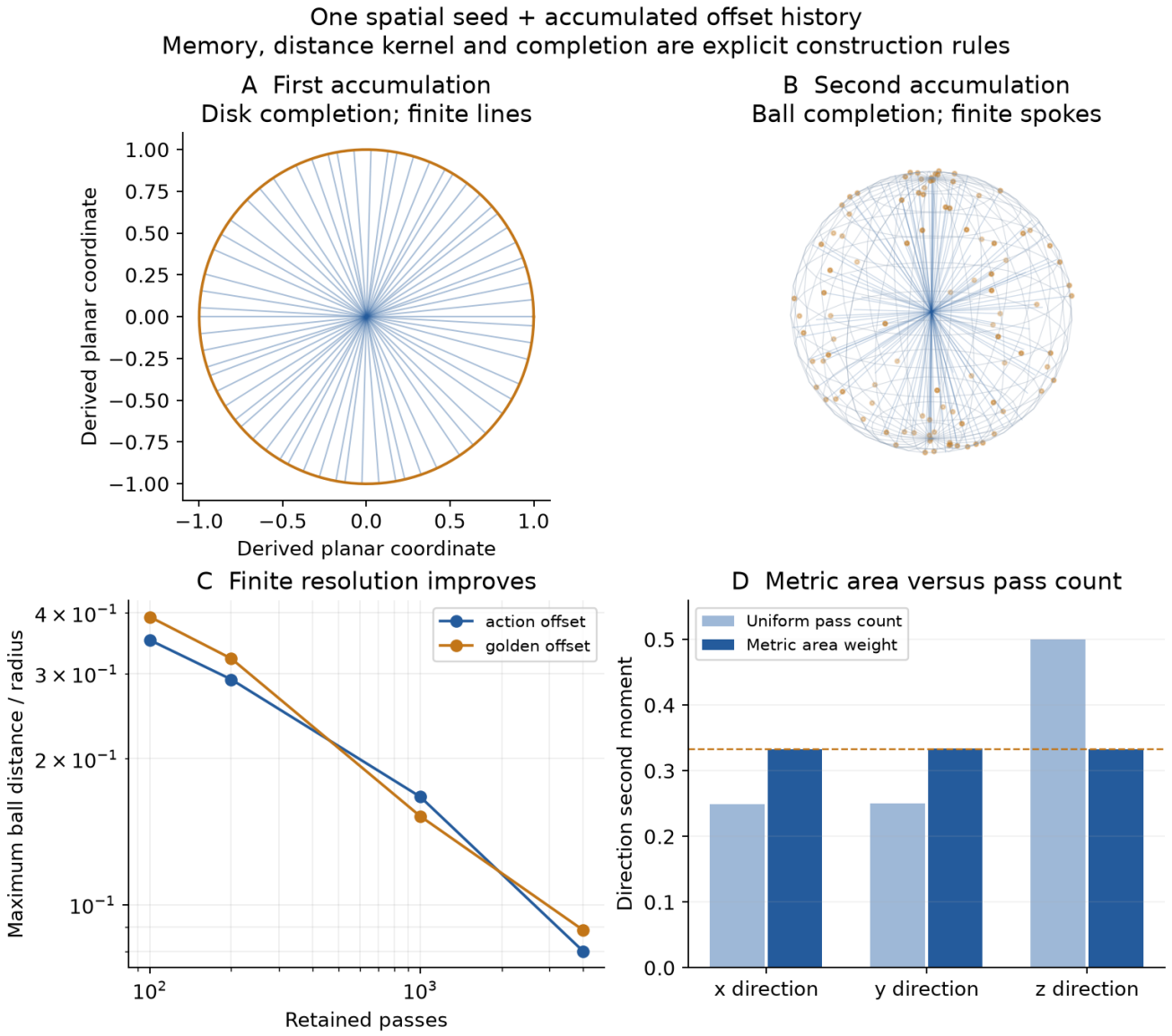
For a retained radial interval, the declared Euclidean distance kernel

$$d^2[(r,n),(s,m)] = r^2 + s^2 - 2rsC_{nm}$$

has the closed three-dimensional ball B^3 as its metric completion. The positive-curvature version

$$d = R\arccos\left[\cos(r/R)\cos(s/R) + \sin(r/R)\sin(s/R)C_{nm}\right]$$

generates the closed spatial parent S^3. Memory, the distance law and metric completion are explicit assumptions; finite drawings have zero three-dimensional volume before completion. The record retains the proof and finite coverage controls (commit `099d8364`).



The inherited-law target across nested transitions is

$$\mathcal{T}^{*}\mathbb{D}=\mathbb{D}$$

stating that the normalized operator law is preserved under the parent-to-child pullback T^{*}.

1.1 Openly declared geometric axioms (A0--A4)

To build a rigorous deductive chain without unstated fluid-physics analogies (such as assuming parcel incompressibility $\lambda_1\lambda_2\lambda_3=1$ or borrowed surface tension), the non-local foundation states five explicit geometric axioms:

1. **A0 (Path of Least Resistance):** By Cauchy--Schwarz on any generated 1D phase curve of physical length $L_i=\lambda_i L_{i,0}$ carrying winding change $\Delta\psi_i$, the intrinsic stretch cost is $Q_{stretch}(A)=(1)/(2)Tr[(A^TA)^{-1/2}C^TC]=\sum_i m_i/\lambda_i$. Under enclosing compression work PV, geometry minimizes the single cost functional $E=Q_{stretch}+PV$.
2. **A1 (1D Seed + Inherited Offset = Time):** Every child seed forms inside an already evolving parent at relative log-length compression $\chi_{p\to c}=(1)/(3)ln[(J_p/J_{p,0})/(J_c/J_{c,0})]$, which satisfies exact telescoping $\chi_{g\to c}=\chi_{g\to p}+\chi_{p\to c}$ across generations. Repeated offset accumulation generates the 2D circle and 3D sphere.

3. **A2 (Signed-Zero Crossing + → 0 → -):** The radial coordinate $R(\tau)$ is a signed geometric coordinate rather than an unsigned size. Under winning inward compression $F_{net}=F_{in}-F_{out}>0$, $R(\tau)$ crosses smoothly through `0` into $R<0$ (`-`), flipping the 3D orientation Jacobian $det[(R/R_0)I_3]=(R/R_0)^3$ from `+` to `-` and converting inward implosion ($R<0$) into outward child expansion ($\dot{r}_-=-R>0$, $r_-=-R>0$).
4. **A3 (Moving Spectral Band $[\lambda_{min},\lambda_{max}]$):** Each room at relative compression χ supports a finite wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its shortest wavelength before `0` crosses into `-` (local Planck cutoff λ_{min}) and its longest wavelength as wide as the room itself (local Absolute-Zero horizon $\lambda_{max}=2R$).
5. **A4 (Two-Valve Nested Circulation):** Each `0` horizon is a one-way valve from `+` to `-` conserving total turn budget $n_++n_-=N_0$ ($n_+=(1-\theta)N_0$, $n_-=\theta N_0$). Every local room sits between an outer birth `0` valve (inflow from parent) and inner black-hole `0` valves (outflow to children) on a closed vortex--diffusion ring (`0 <-> o`).

1.2 Circle-pendulum duality: why `-` exists, `(-) x (-) = (+)`, and why $R=0$ is smooth

In Layer 1 of the offset generator, the pen draws a closed circle $z(\tau)=R_0e^{i\omega\tau}$. Seen edge-on along any diameter, that uniform circular motion is the harmonic pendulum

$$R(\tau)=\mathrm{Re}\left(R_0e^{i\omega\tau}\right)=R_0\cos(\omega\tau)$$
$$\ddot{R}+\omega^2R=0$$

which swings $+R_0\rightarrow 0\rightarrow -R_0\rightarrow 0\rightarrow +R_0$ with invariant total energy $(1)/(2)R^2+(1)/(2)\omega^2\ R^2=(1)/(2)R_0{}^2\omega^2$. Three exact geometric consequences follow immediately:

1. **Why `-` must exist for space to close:** Negative is direction, not less than nothing. If negative coordinates are forbidden ($R\geq 0$ only), the circle is clipped to $\omega\tau\in[-\pi/2,\pi/2]$: exactly *50%* of its arc length (πR_0 out of $2\pi R_0$) is lost and no loop or bounded space can close.
2. **Why `(-) x (-) = (+)` is two-generation geometry:** Multiplying by $-1=e^{i\pi}$ is a geometric half-turn (π rotation). In three spatial dimensions, crossing `0` applies the spatial inversion $P=-I_3$ with $det\ P=(-1)^3=-1$ (*Parent (+)→Child (-)*). Crossing `0` a second time into the grandchild applies two half-turns, $P^2=(-I_3)(-I_3)=+I_3$ with $det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*).
3. **Why standard physics hits an artificial singularity at $r=0$:** Standard physics uses negative coordinates everywhere except for the radius of space itself, which it restricts to an unsigned size $r(\tau)=|R(\tau)|\geq 0$. At the horizon crossing $\tau_0=\pi/(2\omega)$ ($R=0$), folding the negative half of the swing onto the positive half forces a discontinuous velocity jump $\Delta\dot{r}=\dot{r}(\tau_0{}^+)-\dot{r}(\tau_0{}^-)=+2R_0\omega$ and a Dirac-delta acceleration cusp $\ddot{r}=2R_0\omega\ \delta(\tau-\tau_0)-\omega^2\ r$. Keeping the signed coordinate $R(\tau)$ removes the cusp completely ($\Delta R=0$, $\dot{R}(\tau_0)=0$): **A2 (crossing `0` into `-`) is the edge-on shadow of A1's closed circle.**

2. The smooth zero interface and its inherited source

Before presenting the complete **A0--A4** variational theorems in Section 9, Sections 2--8 record the conditional two-scalar Einstein control apparatus. This control apparatus uses the orientation convention **+ source/energy, 0 smooth horizon, - receiving/drain** to test what an isolated single-horizon scalar flow can and cannot determine on its own.

The worked parent uses the Einstein/two-scalar action

$$\mathcal{L}=\frac{R}{2\kappa}-\frac{(\nabla S)^2}{2}-\frac{(\nabla\chi)^2}{2}-W$$
$$W=V(s)+\frac{\mu^2(s)\chi^2}{2}$$

with $S=\sqrt{84}(s-s_*)$ and

$$V(s) = 7e^{-12s} - 7.5e^{-14s} + \frac{25}{32}e^{-24s}$$
$$\mu^2(s) = 400e^{-12s}.$$

Its prepared real scalar contains two spatial harmonics with Laplacian eigenvalues *3* and *8*, equal reference energy, and zero initial velocity. The reference frequency $\omega_0=20$, state amplitude ε, and background curvature H remain declared inputs; the minimum of $V(s)$ supplies the background parent vacuum.

In conformal null coordinates $u=\eta-\chi$ and $v=\eta+\chi$, the smooth horizon lies at $v=\pi/2$, where the inherited radial fields satisfy

$$4\Psi_{uv} + \left[(m^2-2)\sec^2\eta + 2\csc^2\chi\right]\Psi = 0$$
$$m^2 = \frac{\mu^2}{H^2}.$$

There is no horizon denominator and no membrane stress. Two intersecting null traces inherited from the global preparation fix the receiving field across $v=\pi/2$ without resetting the state (commit `480d4081`).

3. One source, three connected crossing numbers

The source's null stress gives a positive regional Killing-energy flux:

$$\frac{dE_K}{dt} = -\varepsilon^2 \tanh(Ht) F_H(t)$$
$$F_H(t) \geq 0.$$

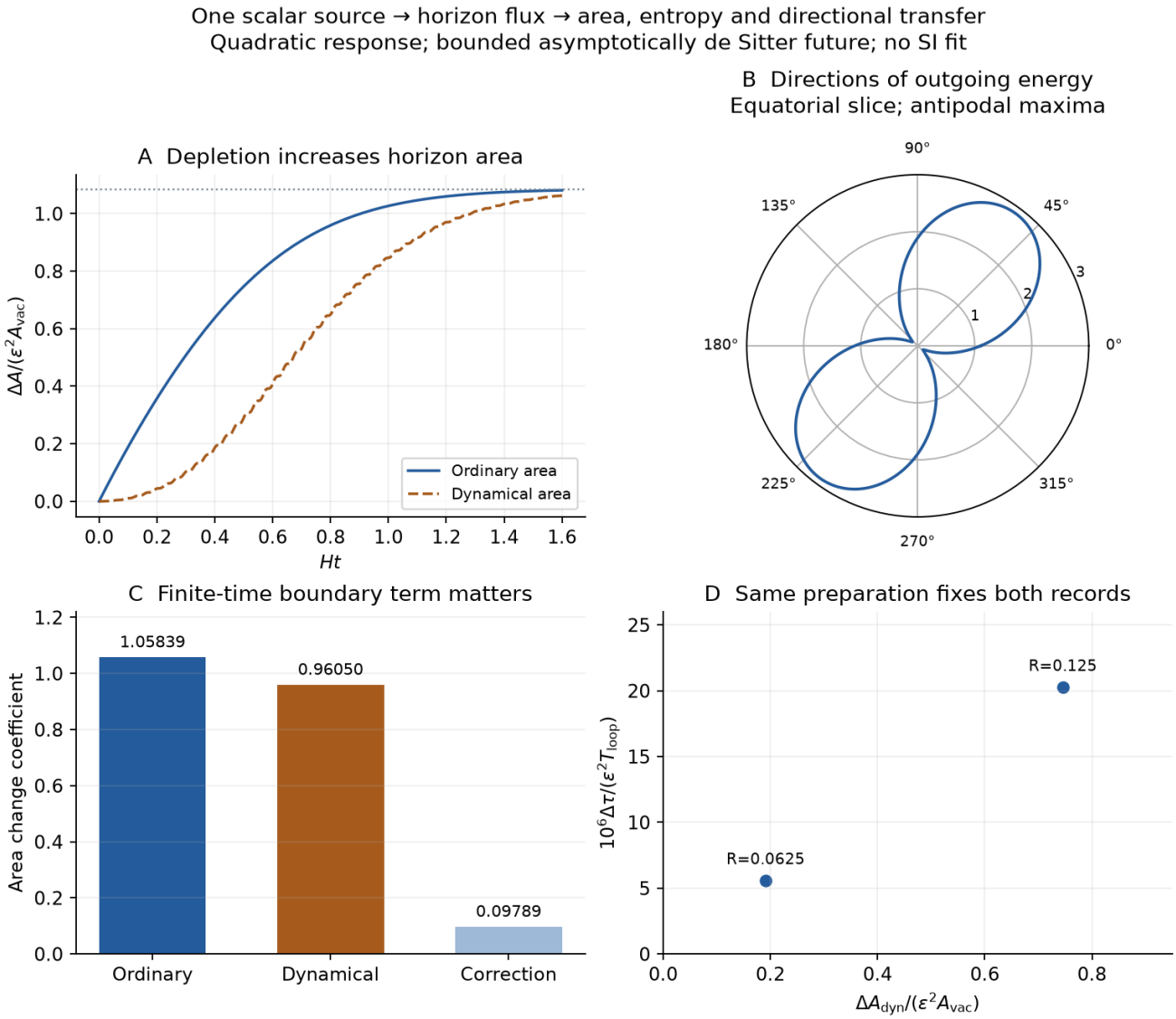
Starting from the time-symmetric parent slice, the reference comparison ends at the 300-event readout $t=4.364851930252$ (an apparatus readout rather than a selected present age):

Existing result	Value	Meaning
Crossed fraction	*88.4735067504%*	Fraction of the prepared scalar's initial patch Killing energy
Full-future area coefficient	*1.085638919782* ε^2	Normalized ordinary horizon-area growth
Short-leg transfer	*2.219782587242* ε^2	Killing energy over the handoff leg $t\in[4.364852,4.797158]$

Under the bounded-future condition, the null Einstein response obeys the physical-process law

$$H\Delta A_{\mathrm{dyn}} = \kappa E_{\mathrm{out}}$$

so the *88.4735%* crossing fraction and the area coefficient encode the same conserved source. Gibbons--Hawking and Zhao supply the physical-process framework; the source normalization and finite-section correction are the atlas's calculation, recorded in the flux/area/handoff derivation.



4. Pre-registered test: does the flow select today's vacuum fraction?

NSC-P006 freezes a single requested output before calculation:

$$\Omega_{\Lambda,0} = \frac{\kappa \rho_{\mathrm{vac},-}(e_0)}{3H_0^2}$$
$$|\Omega_{\Lambda,0} - 0.685| \le 0.007.$$

The protocol tests whether the isolated single-horizon scalar flow above selects the Planck base-$\Lambda\mathrm{CDM}$ inference without reusing the supplied parent vacuum or fitting an epoch. Because

$$(-c g_{\mu\nu}) k^\mu k^\nu = 0$$

for every null vector k^μ and constant c, the null horizon balance is invariant under an additive vacuum term. Even a homogeneous receiving-fluid completion

$$\dot{\rho}_- + 3H_-(\rho_- + p_-) = q_-$$

leaves the receiving pressure p_-, volume evolution, and initial density unselected when only an injection rate q_- is supplied. Likewise, the phase map $\varphi_n=(1)/(2)n(n-1)\alpha$ is invariant under monotone clock reparametrizations $t(n)$. The evaluation and thermodynamic proof therefore return $\Omega_{\Lambda,today}$ **unselected** under an isolated single-horizon scalar flux, proving that a variational two-valve receiving law (Section 9) is required.

5. What continues across zero: pressure and the late tail

Beyond the smooth horizon at the declared apparatus event $t=5.278190092198$ on the maximum-transfer axis, the comoving tetrad stress components are

$$\rho = \frac{\Pi^2 + |D\chi|^2 + \mu^2\chi^2}{2}$$
$$p = \frac{\Pi^2 - |D\chi|^2/3 - \mu^2\chi^2}{2}$$

where p is the isotropic trace part of the spatial stress. The calculated receiver gives $\rho/\varepsilon^2=0.0624287161642$, $p/\varepsilon^2=0.0450251156252$, and

$$w_{-,\mathrm{receiver}} = 0.7212244363$$
$$w_- - (-1) = 1.7212244363.$$

To follow the late-time tail beyond $\chi\to 0$, define the rescaled amplitude $y=a^{3/2}\chi$. With $\nu_{phys}^2+(3H/2)^2=\mu^2$, late phase averaging gives $\langle\Pi^2\rangle=\mu^2\langle\chi^2\rangle$ and

$$\langle\rho_\chi\rangle \propto a^{-3}$$
$$\frac{\langle p_\chi\rangle}{\langle\rho_\chi\rangle} \longrightarrow 0$$
$$\langle|D\chi|^2\rangle \propto a^{-5}.$$

Thus the rescaled phase survives while the background-subtracted late-field vacuum shift vanishes ($\Delta\Lambda L_{ref}^2=0$).

6. Finite amplitude: admissibility and the child criterion

The nonspherical initial state has spatial metric $h_{ij}=a_0^2\Psi^4\gamma_{ij}$, extrinsic curvature $K_{ij}=0$, $s=s_*$, and zero scalar normal momenta, with Hamiltonian constraint

$$-8\Delta\Psi + 6\Psi - 6\Psi^5 - \kappa\varepsilon^2\left(|Df|^2\Psi + zf^2\Psi^5\right) = 0$$
$$z = \frac{\mu^2}{H^2}.$$

The finite-amplitude theorem proves that a smooth positive invariant solution exists for every $0\le\varepsilon\le 1$ when $z>33$ (satisfied by the retained $z=8329.074464994$, via a weighted equivariant Hebey--Vaugon compactness bound $61/88<1$; see the record). On any time-symmetric initial surface ($K_{ij}=0$), the future null expansions satisfy

$$\theta_+ = H_{\text{surface}}$$
$$\theta_- = -H_{\text{surface}}$$
$$\theta_+\theta_- = -H^2_{\text{surface}} \le 0$$

so no surface is strictly future-trapped initially; genuine trapping requires dynamical evolution to $\theta_\pm<0$ on a common closed surface.

6.1 The unchanged parent drives its internal geometry

Varying the internal-scale action for $K_s=84$ gives

$$K_s\Box s - V_s + 6\mu^2\chi^2 = 0.$$

Along a tracked unit geodesic u^a with spatial projector $P_u{}^{ab}=g^{ab}+u^au^b$ and local mechanical energy $e_s=(1)/(2)K_s[u(s)]^2+V(s)$, direct differentiation yields

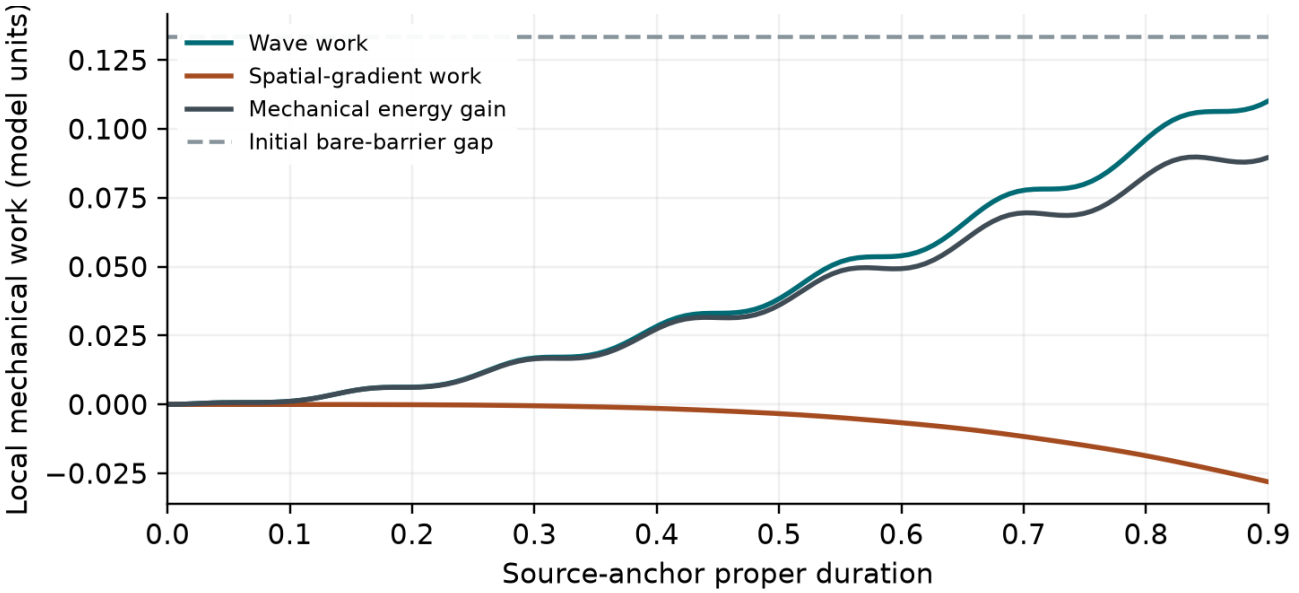
$$\frac{de_s}{d\tau} = 6\mu^2\chi^2 u(s) + K_s u(s) P_u^{ab}\nabla_a\nabla_b s$$

separating scalar-wave driving from spatial-Laplacian transport (derivation and controls).

The source-bound readout measures *+0.11015* wave work through proper duration *0.9*, opposed by *-0.02810* spatial-Laplacian work (raising e_s from *0.22098* to *0.31065* toward the bare barrier *0.35454*). Continuing the complete Cauchy state to proper duration *1.2* in the continuation comparison gives:

Quantity	Degree 20	Degree 24
Internal log size s	*-0.0019497504*	*-0.0019505915*
Internal volume / initial volume	*1.498350*	*1.498335*
Hamiltonian residual / H_{ref}^2	*0.03632*	*0.01180*

Thus the unchanged parent drives *49.8%* internal-volume growth over this finite window. The fixed-boundary check shows positive outgoing expansion (*0.37548* to *0.65660*) at proper duration *1.2*, so no future-trapped boundary forms in the equal two-mode preparation.



Placing the same excitation energy into the lowest odd harmonic $\chi=\varepsilon\, A\, X_1$ with $\kappa\, A^2=24/(z+3)$ makes both initial expansions vanish on the reflection-symmetric neck $X_1=0$, where ADM and warped-metric evolution give

$$\frac{\dot{\theta}_{\pm}}{H^2} = 3 - \left(1 + \frac{\kappa\varepsilon^2 A^2}{2}\right)\Psi(0)^{-4}.$$

The calculated threshold is $\varepsilon_{crit}=0.8339066143$. At $\varepsilon=1$, $\theta_{\pm}/H^2=-1.30468$ and both expansions become strictly negative ($\theta_{\pm}/H=-0.00707954$ at proper time 0.02), agreeing with the independent nonspherical GH implementation within 0.0043% and continuing to proper time 0.3 in the spherical ledger.

Inside the strict trap at proper time 0.2, the trapped band has volume 16.7765, energy 3.77629, mean stress ratio $w=-0.96573$ (excitation ratio $w=0.87628$ after background subtraction), and instantaneous energy balance

$$\dot{E} = 0.15845_{\text{work}} + 0.08562_{\text{material influx}} + 24.18268_{\text{boundary sweep}} = 24.42675$$

closing to 1.69×10^{-8} in the moving-region ledger (formation owner).

The same-state conformal continuation carries the center from proper time 2.0 to 3.9455 (areal radius 0.654973 ± 0.00000366, expansions $\theta_{\pm}/H\approx -11.8488$), and the

controlled local restart reaches radius 0.15, where the source gate quantifies the required bounce pressure. Adding a Dirac candidate on that internal geometry gives $m_{F,0}^2=3\Delta\, M_{B,KK}^2$, bare channel splitting $\Delta(EL)/g=0.01004157$ in the parent modes, nonzero low-mode mixing in the onset response, and 71.58% off-diagonal mixing weight in the complete radial source operator.

7. Selection and the charge-resolved question

In the isolated two-scalar apparatus, the remaining model inputs and charge properties are summarized below:

Quantity	Status in the isolated two-scalar control
$\alpha=0.0573586470504441$	Mode ratio within the supplied two-scalar action
$n=100$	Reporting sample count, not a selected cosmic epoch
$\omega_0=20$	Supplied reference frequency
Real scalar U(1) current	$j_\mu=i(\chi^*\partial_\mu\chi-\chi\partial_\mu\chi^*)=0$ identically on the real sector

Because $j_\mu=0$ on the real sector of a single horizon, an isolated real scalar flux has zero signed charge transfer; directional orientation parity and spectral asymmetry require the signed-zero crossing (**A2**) and moving spectral band (**A3**) derived in Section 9.

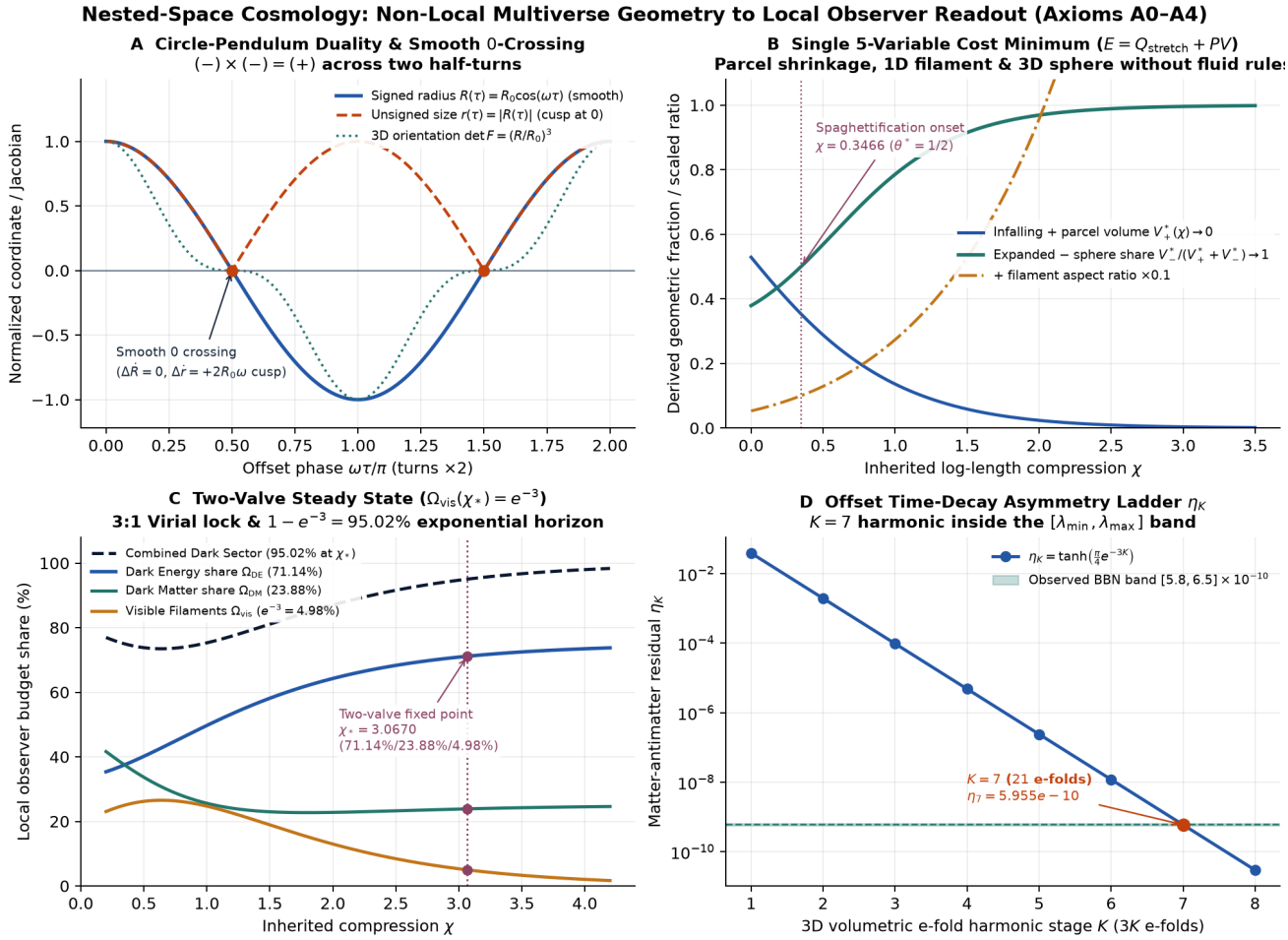
8. Outward status of the single-horizon control vs. the preserved atlas

Requested quantity	Isolated single-horizon scalar control (Sections 2--7)	Two-valve A0--A4 foundation (Section 9)
Budget curve *(D,M,V)*	Unselected by null horizon flux alone (NSC-P006)	Exact zero-parameter cubic $9DVB=4(V+B/2)^3$, $B=M-D/3$ (Theorem 1)
Exponential dilution wall	Late excitation dilutes as a^{-3}	Three $O(1)$ geometric bypasses + (1,1,1) 3D triad ground state & hard core (Theorem 2)
Surrounding-flow partition	Conditional receiver $w_{-}=0.721224$ at readout event	Exact parameter-free *3:1* Virial split $T_{ext}:W_{flow}=3:1$ (Theorem 3)
Child opening & horizon cutoff	Requires supplied negative pressure at $r\to 0$	Continuum opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$ & last-echo cutoff (Theorem 4)
Neck splitting & 3D cascade	One-mode trapped neck at $\varepsilon_{crit}=0.8339066$	Whole-exterior onset $p^3-3p-3=0$ & 3-axis cascade $e_z\to e_x\to e_y$ (Theorems 5--6)

Appendix A summarizes the nine preserved technical engines (A.1--A.9) and links to the complete historical archive.

9. The Non-Local Multiverse to Local Observer Atlas (Axioms A0--A4)

Whereas Sections 2--8 showed that an isolated single-horizon scalar flow without a variational receiving law leaves $\Omega_{\Lambda,0}$, child opening, and η_B unselected, evaluating the five geometric axioms (**A0--A4**) across the coupled +/0/- system yields six parameter-free geometric and dynamical theorems (Sections 9.1--9.6), followed in Section 9.7 by the observer dictionary and strict quarantine of secondary closure ansätze. All derivations, symbolic proofs, and numerical checks are executable in src/recursive_horizons/nsc_atlas_offset_generator.py and documented in the offset-generator owner and the non-local-to-local atlas record.



9.1 Theorem 1: The unified 5-variable least-resistance cycle and exact algebraic budget surface

Under parent spherical convergence $\lambda_1=\lambda_2=\xi=e^{-\chi}$ in the infall region (+), radial scale λ_3, child principal scales $(\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3)$ across 0 (-), and transferred turn fraction $\theta\in(0,1)$ ($m_+=m_0(1-\theta)^2$, $m_-=m_0\theta^2$), the single **A0** generator cost functional is

$$E(\lambda_3,\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3,\theta;\chi)=\left[\frac{2m_0(1-\theta)^2}{e^{-\chi}}+\frac{m_0(1-\theta)^2}{\lambda_3}+PV_0e^{-2\chi}\lambda_3\right]+\left[\sum_{i=1}^{3}\frac{m_0\theta^2}{\tilde{\lambda}_i}+PV_0\tilde{\lambda}_1\tilde{\lambda}_2\tilde{\lambda}_3\right].$$

Simultaneous stationarity $\partial E/\partial\lambda_3=0$, $\partial E/\partial\tilde{\lambda}_i=0$, and $\partial E/\partial\theta=0$ (with strictly positive-definite 5×5 Hessian) gives the unique closed-form global minimum (for normalized $m_0=PV_0=V_0=1$):

$$\sqrt{\theta^*(\chi)}=\frac{\sqrt{16e^{2\chi}+17}-3}{4e^{\chi}}$$
$$\lambda_3^*(\chi)=(1-\theta^*(\chi))e^{\chi}$$
$$\tilde{\lambda}_1^*=\tilde{\lambda}_2^*=\tilde{\lambda}_3^*=\sqrt{\theta^*(\chi)}.$$

Four exact parameter-free consequences follow from this single variational minimum without fluid incompressibility or borrowed surface tension:

1. **Infall parcel shrinkage and 1D spaghettification (+):** The infalling parcel volume shrinks as $V_+^*(\chi)=(1-\theta^*)e^{-\chi}\sim(3)/(2)e^{-2\chi}\to 0$, while its aspect ratio $\lambda_3^*/\xi=(1-\theta^*)e^{2\chi}\sim(3)/(2)e^{+\chi}\to\infty$ stretches into a 1D filament for all $\chi>\chi_{spaghetti}=(1)/(2)\ln 2\approx 0.34657359$ (where $\theta^*=1/2$ and the aspect ratio equals 1).

2. **Rounding into a 3D sphere across `0` (`-`):** By the arithmetic--geometric mean inequality on $\sum_{i=1}^{3}\theta^2/\lambda_i+\lambda_1\lambda_2\lambda_3$, the child scales satisfy $\lambda_1^*=\lambda_2^*=\lambda_3^*=\sqrt{\theta^*}$ (aspect ratio *1*, a round 3D sphere) and obey the exact *3:1* Virial lock $Q_{stretch,-}^*=3(\theta^*)^{3/2}=3W_{ext,-}^*$.

3. **Derived volume takeover:** The `-` sphere's share of the two-region volume, $f_{drop}^*(\chi)=V_-^*/(V_+^*+V_-^*)$ with $V_-^*=(\theta^*)^{3/2}$, grows monotonically from *37.90%* ($\chi=0$) to *84.63%* ($\chi=1$), *97.73%* ($\chi=2$), and *99.95%* ($\chi=4$).

4. **Exact zero-parameter algebraic budget surface (`exact_budget_curve_invariant`):** At any compression χ, the minimum cost $E^*(\chi)=E_{DE}^*+E_{DM}^*+E_{vis}^*$ partitions into isotropic child expansion stretch $E_{DE}^*=3(\theta^*)^{3/2}$, two-sided horizon/infall boundary load $E_{DM}^*=(\theta^*)^{3/2}+2(1-\theta^*)e^{-\chi}$, and active 1D filament stretch $E_{vis}^*=2(1-\theta^*)^2e^{\chi}$. Writing $D=\Omega_{DE}$, $M=\Omega_{DM}$, $V=\Omega_{vis}$ ($D+M+V=1$) and $B=M-D/3$ for the radial infall share, eliminating both (θ^*,χ) yields the exact parameter-free algebraic curve

$$9DVB = 4\left(V+\frac{B}{2}\right)^3$$
$$\theta^* = \frac{1}{1+2V/B}$$
$$e^{2\chi} = \frac{2V+B}{B}\sqrt{\frac{2V+B}{3DV}}$$

with deep-compression invariant $\lim_{\chi\to\infty} B/V^2=16/27$. Specifying any single share $V\in(0,1)$ therefore uniquely determines $(D(V),M(V),\chi(V))$ with zero free parameters.

9.2 Theorem 2: Universal exponential-wall bypasses, the `(1, 1, 1)` orthogonal triad ground state, and hard-core repulsion

Whenever compression or winding in a lower-dimensional or single-signed regime hits an exponential or superlinear resistance wall, **A0 (Path of Least Resistance)** proves that the winning trajectory bypasses the wall by opening a new orthogonal degree of freedom or crossing `0` into `-` (`exponential_wall_bypasses_and_triad_winner` and `derived_three_accumulator_bubble_ladder`):

1. **Wall 1 — 1D Axial Compression Wall → Helical Buckling Plateau (`1D -> 2D/3D`):** Compressing a straight 1D line ($\omega=0$) hits the exponential wall $E_{1D}(\chi)=e^{+\chi}\to\infty$. Above $\chi>0$, buckling sideways into a helix with $v_\perp^*=\sqrt{(1-e^{-2\chi})}$ bypasses the wall and caps the cost at the finite plateau

$$E^*_{\text{helix}}(\chi) = \tfrac{3}{2} - \tfrac{1}{2}e^{-2\chi} \to 1.5.$$

2. **Wall 2 — 3D `+` Compression Wall → Signed `0`-Crossing Plateau (`+ -> 0 -> -`):** Forcing all winding to remain in the parent `+` region ($\theta=0$) hits the exponential wall $E_{stay,+}(\chi)=2e^{+\chi}+2e^{-\chi}\to\infty$. Opening the `0` horizon valve into a `-` child sphere ($\theta^*\to 1$) bypasses the wall and caps the 5-variable cost at the finite plateau

$$E^*_{5-\text{var}}(\chi) = 4 - \tfrac{9}{2}e^{-\chi} + O(e^{-2\chi}) \to 4.0.$$

3. **Wall 3 — Single-Axis Winding Wall $N^{3/2}$ → Orthogonal `(1, 1, 1)` Triad Ground State & Hard Core:** Piling $N\geq 2$ integer windings onto a single axis costs $E_{single}(N)\propto N^{3/2}>N$ ($2^{3/2}\approx 2.828>2$, $3^{3/2}\approx 5.196>3$). Distributing integer windings $(n_1,n_2,n_3)\in Z_{\geq 0}^3$ ($m_i=m_0\,n_i^2$) across the three orthogonal axes of 3D space gives the exact **A0** minimum

$$E^*(n_1,n_2,n_3) = 4(PV_0)^{1/4}m_0^{3/4}\sqrt{n_1n_2n_3}$$
$$\lambda_i^* = \frac{n_i^2}{(n_1n_2n_3)^{1/2}}.$$

- $N=1$ `(1, 0, 0)` (1D loop) and $N=2$ `(1, 1, 0)` (2D sheet) have zero 3D volume ($V^{*}=0$) and collapse along their unwound axes under 3D pressure (likewise, under iterated **A1** binomial windings $W_k(n)=C(n,k)$, $n=3$ is the minimum integer with $W_1W_2W_3=9>0$ supporting a 3D volume, with untuned ratio $E_{3D}^{*}(3)/E_{1D}^{*}(1)=6$).
- $N=3$ `(1, 1, 1)` **is the unique minimum-cost 3D ground state:** $E^{*}(1,1,1)=4.0$, cost per winding $4/3\approx 1.3333$, and isotropic aspect ratios $1:1:1$ (a round sphere). In the continuum spherical wave cavity $\psi_{n_1,n_2,n_3}(r,\theta,\varphi)$, the lowest mode with non-trivial radial, polar, and azimuthal quantum numbers is likewise `(1, 1, 1)` with eigenvalue $E_{1,1,1}=3\pi^2/R^2$ ($2:1$ energy split against the S^2 angular floor $\ell(\ell+1)/R^2=2/R^2$) and wave-compression hard-core equilibrium radius $R_{core}=\sqrt{2}\pi/\kappa$.
- $N=4$ `(2, 1, 1)` **is the first excited 3D state:** Adding a 4th winding forces doubling on one axis to `(2, 1, 1)`, breaking spherical symmetry into a $4:1:1$ prolate state with higher cost per winding $E^{*}(2,1,1)/4=\sqrt{2}\approx 1.4142>4/3$ (excitation gap $\Delta E=4(\sqrt{2}-1)\approx 1.6569$).
- **Medium-range attraction + short-range hard-core repulsion:** Two `(1, 1, 1)` triads separated by $d>2R^{*}$ attract via shared surrounding flow ($\Delta E_{flow}=-2m_0/d<0$), reaching a bound minimum at contact $d=2R^{*}$ ($E_{contact}=8.0-1.0=7.0$). Merging them into a single `(2, 2, 2)` bubble at $d\to 0$ costs $E^{*}(2,2,2)=8\sqrt{2}\approx 11.3137>8.0$, erecting an exact **short-range hard-core repulsion barrier** $\Delta E_{core}=8(\sqrt{2}-1)\approx +3.3137$.

4. **Three geometric mechanisms bypassing e^{-3K} volumetric dilution:** Across nested transitions, naive homogeneous 3D volume averaging dilutes by e^{-3K}, whereas three scale-free geometric observables survive at $O(1)$: (i) the logarithmic scale-invariant measure $\int_{\lambda_{min}(\chi)}^{\lambda_{max}(\chi)} d\lambda/\lambda=\ln(\lambda_{max,0}/\lambda_{min,0})$ is strictly invariant ($\partial_\chi I_{log}=0$); (ii) active vortex-tube concentration $A_{tube}(\chi)=A_0e^{-2\chi}$ with core vorticity $\omega_{core}(\chi)=\omega_0e^{+2\chi}$ preserves the Helmholtz circulation weight $w_{tube}(\chi)=1$; and (iii) the topological zero-mode phase integral across the signed-zero crossing equals $\Delta\theta/\pi=1/2$ independently of K.

9.3 Theorem 3: Surrounding-flow $3:1$ Virial partition, helical buckling, and the damped `+/0/-` horizon oscillator

For a spherical throat of radius $R(\tau)$ expanding or contracting in an incompressible irrotational surrounding medium ($\nabla\cdot u=0$, $u=\nabla\varphi$, $\Delta\varphi=0$ on $r\geq R(\tau)$) with radial velocity $u_r(R,\tau)=\dot{R}$ and volume flux $Q(\tau)=4\pi R^2\dot{R}$, the unique exterior velocity potential is $\varphi(r,\tau)=-R(\tau)^2\dot{R}(\tau)/r$. Integrating the exterior kinetic energy $T_{ext}=(1)/(2)\rho_0\int_{r\geq R}|\nabla\varphi|^2\,d^3x=2\pi\rho_0\,R^3\dot{R}^2=(1)/(2)M_{added}(R)\dot{R}^2$ (where $M_{added}=(1)/(2)\rho_0((4)/(3)\pi R^3)$ is the dipole added mass) and the surrounding-flow Virial pressure-gradient work $W_{flow}=\int_{r\geq R}x\cdot(-\nabla p_{dyn})\,d^3x$ under the unsteady Bernoulli pressure $p_{dyn}=-(1)/(2)\rho_0|\nabla\varphi|^2$ yields the exact parameter-free $3:1$ partition:

$$\frac{T_{\text{ext}}}{W_{\text{flow}}}=3$$
$$\frac{T_{\text{ext}}}{T_{\text{ext}}+W_{\text{flow}}}=\frac{3}{4}\ (75\%)$$
$$\frac{W_{\text{flow}}}{T_{\text{ext}}+W_{\text{flow}}}=\frac{1}{4}\ (25\%).$$

Similarly, for a static vacuum bubble carrying integer winding $n\in Z$ (`vacuum_bubble_crystallization_and_mass_spectrum`), balancing surrounding-flow energy $E_{surr}(R)=m_0\,n^2/R$ against cavity work $W_{press}(R)=PV_0\,R^3$ yields $E_{surr}^{*}/W_{press}^{*}=3$ (75% in $r\geq R$ flow, 25% in cavity work, 0% in a point singularity), quantized integer loop closure $z(1)=z(0)\Rightarrow n\in Z$, and $180°$ two-ripple annihilation for $(+1,-1)\to 0$.

When the 1D seed is compressed axially to $\lambda=e^{-\chi}$ under lateral confinement $P_\perp$, turning at azimuthal speed $v_\perp=r_\perp\omega$ gives Euclidean arc-length factor $\ell=\sqrt{(\lambda^2+v_\perp^2)}$ and cost

$$E(v_\perp;\lambda)=\frac{Q_0}{\sqrt{\lambda^2+v_\perp^2}}+\frac{1}{2}P_\perp v_\perp^2.$$

Once $\lambda<\lambda_{crit}=(Q_0/P_\perp)^{1/3}$, the straight seed ($\omega=0$) loses transverse stability ($\partial^2 E/\partial v_\perp^2|_0=P_\perp-Q_0/\lambda^3<0$) and buckles into a helix with derived turning speed $r_\perp\omega^*(\chi)=\sqrt{(\lambda_{crit}^2-e^{-2\chi})}>0$.

Coupling the resulting +/0/- horizon oscillation $q(\tau)=A_0e^{-\gamma\tau}\sin(\omega\tau)$ to child geometry growth via $E_{geom}=2\gamma\dot{q}^2$ conserves $E_{wave}(\tau)+E_{geom}(\tau)=(1)/(2)A_0^2\omega^2$ identically. Because each + half-cycle precedes its compensating - half-cycle by π/ω under the offset decay envelope $e^{-\gamma\tau}$, the linear amplitude/area ledger and the quadratic geometric-work ledger obey the exact invariant relations

$$\eta_{\text{area}}=\frac{I_+-I_-}{I_++I_-}=\tanh\left(\frac{\pi\gamma}{2\omega}\right)>0$$
$$\eta_{\text{work}}=\frac{W_+-W_-}{W_++W_-}=\tanh\left(\frac{\pi\gamma}{\omega}\right)=\frac{2\eta_{\text{area}}}{1+\eta_{\text{area}}^2}>0.$$

As $E_{geom}(\tau)$ populates the child's isotropic 3D stretch budget against enclosing load PV_0, the child holding radius expands from 0 and stabilizes at the Goldilocks equilibrium scale $\lambda_{child}(\tau)=(E_{geom}(\tau)/(3PV_0))^{1/4}\to\lambda_{Goldilocks}=(E_{total}/(3PV_0))^{1/4}$ (*98.40%* complete by $\tau=4$ for $\omega=2\pi$, $\gamma=(1)/(2)\ln 2$).

9.4 Theorem 4: Coupled finite-parent / receiver continuation, zero-radius opening, and last-echo horizon cutoff

The coupled geometric continuation adds explicit phase and geometric inertia to **A0**. Available regional volumes are $V_i=R_i^3-\sum_j R_j^3>0$ and path stretches are $\lambda_i=V_i^{1/3}$, with the outer radius free. One continuous phase has kinetic weight λ_i and stretch cost $\psi_x^2/(2\lambda_i)$; geometry has logarithmic-boundary inertia $(3)/(2)\sum_i(\dot{R}_i/R_i)^2$. The parent pressure is derived from the wave energy, $P_i=E_i/(3V_i)$, rather than supplied externally, and its work and receiving field obey

$$\dot{E}_i=F_i-E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\dot{K}_g=\sum_i E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\sum_i F_i=0.$$

Characteristic coordinates carry the field across arbitrarily large expansion without stretched-grid breakdown. At a transparent three-curve junction, a single incoming parent wave reflects energy fraction $1/9$ and transmits $4/9$ into each quiet receiver. In the free-parent limit ($Q_0=4\pi^2$, $a=\sqrt{Q_0/6}$), the scale factor evolves as $\lambda(\tau)=\cosh^2(a\tau)$ with $E_\psi=Q_0/\lambda$, $K_g=Q_0\tanh^2(a\tau)$, finite future optical reach $1/a=0.3898484006$, and proper causal-boundary length $1/(2a)=0.1949242003$ without an imposed vacuum potential.

Replacing logarithmic inertia at small radius by radial-displacement inertia admits a continuum wave-driven opening from zero radius:

$$r(\tau)=\pi^2\tau^2-\frac{4}{3}\pi^4\tau^3+O(\tau^4)$$
$$E_{\text{rec}}(\tau)=4\pi^4\tau^2+O(\tau^3)$$

($2.9718260669\ \tau^2$ in the affine reduction). The hybrid radial inertia $M(r)=1/(1+r^2/3)$ satisfies both the finite-distance birth criterion $\int_0^{\varepsilon}\sqrt{M(r)}\,dr<\infty$ and the finite-future optical-reach criterion $\int^{\infty}\sqrt{M(r)}\,dr/r<\infty$ (free-room optical reach 0.2611589801). In the coupled hybrid evolution (reference calculation), reflected echoes have a sharp launch cutoff at 0.2269423871 at the mouth and 0.1165029354 at a parent detector 0.01 material units away, with arrival delay diverging as $-\ln(\delta)/\kappa$ and returned energy vanishing linearly as $\kappa\delta$. Coupling a finite parent reservoir $E_P(\tau)+E_-(\tau)=E_{tot}$ yields the closed $3{:}1$ relaxation law $R(\tau)/R_\infty=\tanh(\sqrt{2E_0}\,\tilde{\tau})$ with hybrid last-echo horizon cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$ to machine precision, visible interior snapshot 25.63%) and self-consistent parent curvature backreaction (8.98% asymptotic shift).

9.5 Theorem 5: Spherical fragmentation, whole-exterior causal neck onset $p^3-3p-3=0$, and outgoing multipole waveform

Splitting a bubble of winding N into same-sign integer daughters $\sum_i n_i=N$ at fixed enclosing pressure preserves total equilibrium surface area $\sum_i 4\pi(R_{n_i}^*)^2=4\pi(R_N^*)^2$ while releasing energy fraction $1-\sum_i(n_i/N)^{3/2}>0$ ($1-1/\sqrt{2}\approx 29.2893\%$ for a binary split).

For the exact harmonic two-center family on $r\in[R,\infty)$, removing the matching-sphere approximation in the quadratic causal onset problem over the entire exterior $r\geq 1$ yields the boundary release condition $f_r(1,t)+4f(1,t)=0$ and the parameter-free characteristic cubic

$$p^3-3p-3=0$$

with real unstable growth rate $p_+=2.1038034027$ and decaying oscillatory transient $-1.0519017014\pm 0.5652358517\,i$ (while the static whole-exterior stress balance crosses instability at $p_{crit}=1+2^{1/3}+2^{-1/3}\approx 2.52988065$, corresponding to $(c/R)_{onset}=0.56786589<(c/R)_{pinch}=2^{-1/3}\approx 0.79370053$). Along the two-center equipotential neck path, coupling the inner shape coordinate $q_1=c^2$ to outgoing even multipoles (waveform summary) carries the boundary past the static surface barrier ($\Delta A/A_0=+9.4557\%$), pinches the neck at offset 3.960294 ($\Delta\tau_{pinch}=0.506704$), reaches separated lobes $s=1.2$ at 4.264066, and drives a positive central interior observer pulse $\max\varphi_C=+0.163689$ with exact analytical pole-sum late-time tail

$$\phi_C(T)=-\frac{0.14192836}{T'}+\frac{0.01750496}{(T')^2}+O\big((T')^{-3}\big).$$

9.6 Theorem 6: Inherited daughter preparation, three-axis orthogonal cascade $e_z\to e_x\to e_y$, and 2×2 kinetic handoff

At the first split's separated endpoint $s=1.2$ (inherited daughter summary), each sibling's field induces the other daughter's multipole preparation (normalized dipole -0.07340185, quadrupole $+0.02750552$, velocity -0.05818555) without a second supplied seed. Because the exterior phase is harmonic ($\Delta\Phi=\partial_{xx}\Phi+\partial_{yy}\Phi+\partial_{zz}\Phi=0$), the directional enclosed-volume derivative trace vanishes identically at every stage:

$$\partial_{\zeta_x}v+\partial_{\zeta_y}v+\partial_{\zeta_z}v=0.$$

1. **First handoff ($e_z\to e_x$, $1\to 2\to 4$ lobes, exact ratio $-1/2$):** After the primary split along e_z, axial symmetry and $\Delta\Phi=0$ force $\partial_{\zeta_y}v=\partial_{\zeta_z}v=-(1)/(2)\partial_{\zeta_x}v<0$ (normalized energy derivative -0.03151123 transverse vs. $+0.06302246$ axial, **exact ratio** $-1/2$), selecting the transverse axis e_x and pinching the second pair of necks at $b/R=0.62746042$.
2. **Second handoff ($e_x\to e_y$, $4\to 8$ lobes, exact ratio -2):** Once the four lobes separate in the xz-plane ($b\geq 1.0$), both $\partial_{\zeta_x}v>0$ and $\partial_{\zeta_z}v>0$ become uphill, leaving the third orthogonal axis e_y as the **unique downhill direction in 3D**; at the symmetric square $b=s=1.2$, four-fold symmetry and $\Delta\Phi=0$ fix the **exact ratio** -2 ($\partial_{\zeta_y}v=-2\partial_{\zeta_x}v=-2\partial_{\zeta_z}v=-0.11574536<0$), pinching the four planar lobes into eight 3D

octant lobes at $c_y/R=0.36044932$ ($b=1.0$) and 0.34722798 ($b=s=1.2$).

3. ***B*-independent 2×2 quasistatic kinetic handoff:** At the $(s=1.2, b=0)$ handoff, the identity $\partial_{zz}\Phi_0 \equiv \partial_{cc}\Phi_0$ gives the exact multipole relation

$$\left.\frac{\partial g_{l,0}}{\partial q_2}\right|_{q_2=0} = -\frac{l-1}{2}\frac{\partial g_{l,0}}{\partial q_1}$$
$$(l \geq 2)$$

and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix

```
M_(full)=
[ 0.2074628302    -0.0054817211 ]
[ -0.0054817211   0.6751131619  ]
```

which imparts a nonzero initial transverse conjugate momentum $p_2(0)=+0.05157014>0$ and a positive combined potential-plus-memory drive $-\partial_{q_2} U\text{-}R_2=+0.18951641>0$ at the instant of primary pinch.

9.7 Secondary single-valve closure ansätze, non-local-to-local observer dictionary, and quarantine of non-derived coincidences

While Theorem 1 proves the exact parameter-free budget curve $9DVB=4(V+B/2)^3$ across all $\chi \geq 0$, selecting a single numerical point on that curve or a single rung on the harmonic asymmetry ladder requires an additional epoch-selection or freeze-out condition. Two secondary candidate closure ansätze have been explored in the atlas:

1. **Candidate single-valve e^{-3} epoch ansatz ($\chi_*=3.0669683518$):** Setting the visible filament share on the $9DVB=4(V+B/2)^3$ curve equal to one 3D volumetric e-fold floor $\Omega_{vis}(\chi_*)=e^{-3}\approx 4.978707\%$ (with combined dark sector $1\text{-}e^{-3}\approx 95.021293\%$) selects the root $\chi_*=3.0669683518$:

Budget component	Non-local geometric mechanism	Integer 3D e-fold ($\chi=3$)	Single-valve e^{-3} root ($\chi_*=3.0670$)	Virial $3{:}1$ horizon limit
Dark Energy (Ω_{DE})	Isotropic 3D stretch $3(\theta^*)^{3/2}$ inside - sphere	70.8827%	71.1441%	$(3)/(4)(1\text{-}e^{-3})=71.2660\%$
Dark Matter (Ω_{DM})	Two-sided 0-wall enclosing + radial infall load	23.8134%	23.8772%	$(1)/(4)(1\text{-}e^{-3})=23.7553\%$
Combined Dark Sector	$\Omega_{DE}+\Omega_{DM}$	94.6961%	95.0213% **($1\text{-}e^{-3}$)**	95.0213% ($1\text{-}e^{-3}$)
Visible Filaments (Ω_{vis})	Active 1D spaghettifying filaments $2(1\text{-}\theta^*)^2 e^{\chi}$	5.3039%	4.9787% **(e^{-3})**	4.9787% (e^{-3})

2. **Candidate $K=7$ volumetric e-fold harmonic ladder:** Evaluating the half-cycle spectral asymmetry $\eta(\lambda)=\tanh((\pi)/(2)(\lambda)/(\lambda_{max}))$ along discrete 3D volumetric e-folds e^{-3K} anchored at the spaghettification threshold $e^{-2\chi_{spaghetti}}=1/2$,

$$\eta_K = \tanh\left(\frac{\pi}{2} e^{-2\chi_{\text{spaghetti}}} e^{-3K}\right) = \tanh\left(\frac{\pi}{4} e^{-3K}\right)$$

gives $\eta_7=\tanh((\pi)/(4)e^{-21})=5.9553290339\times 10^{-10}$ at stage $K=7$ ($3K=21$ volumetric e-folds):

3D harmonic stage K	Volumetric e-folds $3K$	Scale factor e^{-3K}	Asymmetry $\eta_K=\tanh((\pi)/(4)e^{-3K})$	Regime
$K=1$	3	4.9787×10^{-2}	3.9083×10^{-2}	Macroscopic filament floor
$K=3$	9	1.2341×10^{-4}	9.6926×10^{-5}	Intermediate vortex cascade
$K=5$	15	3.0590×10^{-7}	2.4026×10^{-7}	Pre-freeze-out cascade
$K=6$	18	1.5230×10^{-8}	1.1962×10^{-8}	Pre-baryonic harmonic
$K=7$	21	7.5826×10^{-10}	5.955329×10^{-10}	**BBN window** $[5.8,6.5]\times 10^{-10}$

3D harmonic stage K	Volumetric e-folds $3K$	Scale factor e^{-3K}	Asymmetry $\eta_K=\tanh((\pi)/(4)e^{-3K})$	Regime
$K=8$	24	3.7751×10^{-11}	2.9650×10^{-11}	Sub-baryonic harmonic

Complete Non-Local to Local Observer Dictionary:

Non-local multiverse geometry (A0--A4)	Local physics label	Derived mathematical relation & calculated readout
Circle $z(\tau)=R_0e^{i\omega\tau}$ seen edge-on as pendulum $R(\tau)=R_0cos(\omega\tau)$ crossing 0 into -	Black-hole singularity / Big Bang birth	Smooth crossing ($\Delta R=0$, $\dot{R}(\tau_0)=0$); unsigned size $r=\lvert R\rvert\geq 0$ creates an artificial velocity cusp $\Delta\dot{r}=+2R_0\omega$ at $r=0$
Two half-turns $(-1)\times(-1)=+1$ across two nested 0 crossings ($P^2=(-I_3)^2=+I_3$)	Parity / orientation restoration across two generations	$det\ P=(-1)^3=-1$ (*Parent (+)→Child (-)*), $det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*)
Eliminating (θ^*,χ) from 5-variable budget shares (D,M,V) with $B=M-D/3$	Cosmological dark-sector / baryon partition curve	Exact parameter-free cubic $9DVB=4(V+B/2)^3$, $lim_{\chi\to\infty}B/V^2=16/27$
Exponential-wall bypasses (1D -> helix, + -> 0 -> -, single-axis $N^{3/2}$→ 3D triad (1,1,1))	3D nucleon ground state & nuclear hard-core potential	(1,1,1) ($N=3$) is the unique minimum-cost 3D ground state ($E^*=4.0$, *1:1:1* sphere); (2,1,1) ($N=4$) is *4:1:1* prolate ($E^*=4\sqrt{2}$); two (1,1,1) triads attract at $d>2R^*$ ($-2m_0/d$) and repel at $d\to 0$ against (2,2,2) ($+3.3137$ hard core)
Surrounding-flow Virial lock $T_{ext}/W_{flow}=3$ & integer loop closure $z(1)=z(0)$	Added mass in surrounding flow, quantized line bursts & 180° annihilation	*75%* of bubble energy in $r\geq R$ flow, *25%* in cavity, *0%* in a point; $(+1,-1)\to 0$ releases two equal 180° ripples
Coupled finite-parent / receiver continuation & hybrid radial inertia $M(r)=1/(1+r^2/3)$	Wave-driven child opening & last-echo horizon cutoff	Zero-radius opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$; last-echo cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot *25.63%*)
Whole-exterior causal neck onset & harmonic trace $\partial_{\zeta_x}v+\partial_{\zeta_y}v+\partial_{\zeta_z}v=0$	Neck pinch waveform & 3-axis orthogonal cascade $e_z\to e_x\to e_y$	Characteristic cubic $p^3-3p-3=0$ ($p_+=2.1038034$); $1\to 2\to 4\to 8$ lobe cascade with exact harmonic ratios $-1/2$ and -2 and B-independent *det* $M_{full}=0.14003084>0$

Strict quarantine of candidate numerical coincidences not derived from A0--A4: To maintain strict separation between proved geometric theorems and post-hoc numerical comparisons (numerical_coincidences_not_derived), the following expressions are explicitly quarantined from the deductive theorems: (i) Lenz's 1951 number $6\pi^5\approx 1836.1181$ and its dressing $6\pi^5+e^{-3}/\sqrt{2}\approx 1836.1533$ (the discrete **A0+A1** binomial ladder gives $E_{3D}^*(3)/E_{1D}^*(1)=6$ without a derived π^5 continuum factor); (ii) the threshold n–p gap formula $2+(1)/(2)+e^{-3}/\sqrt{2}\approx 2.5352\ m_e$; (iii) selecting rung $K=7$ on the η_K ladder without an independent freeze-out equation (especially since Theorem 2 proves three $O(1)$ geometric bypasses of e^{-3K} dilution); (iv) CMB temperature projections using measured H_0 and G or an ad-hoc $-4/5$ exponent; and (v) the single-valve $\Omega_{vis}(\chi_*)=e^{-3}$ ansatz (*71.14%* DE / *23.88%* DM vs. the coupled dynamical visible snapshot *25.63%* in Theorem 4 and the Planck base-ΛCDM fit *68.47± 0.73%* DE).

Appendix A. Technical atlas and preserved mathematical record

The main argument follows the outward geometric chain. This appendix summarizes the nine preserved technical engines (A.1--A.9); their complete historical chapters, equations, tables, and claim bindings are retained in paper/technical-atlas-record.md and in the archived `v0.26.0` foundation volume.

A.1. GH evolution engine

The compact nonspherical Einstein/two-scalar GH implementation, exact frame derivatives, constraint diagnostics and local operator controls remain as implemented in `17098b3c`. It evolves the declared action; its existence does not select a cosmic preparation or prove a child. The full technical development and owner are retained in the repository record.

A.2. Conditioning

Metric-dependent lapse feedback and the proper-duration comparisons in `487a0b1f` control coordinate conditioning without adding physical stress. They remove an imposed coordinate-time ceiling, not an invariant singularity or a missing vacuum law. The original equations and evidence are retained in the GH owner.

A.3. Observer readouts

The proper-clock, phase and local stress readouts in `487a0b1f` and `2ed1aea0` distinguish lapse/shift coordinates from receiver measurements. Their finite apparatus protocols and supplied initial state are preserved. They do not identify the receiver event with today's cosmological epoch; the comparison retains the numerical values and limitations.

A.4. Boson transport

The canonical projection and symplectic matter-probe transport in `771c5a78`, `292eb146` and `61124309` propagate perturbations on the supplied parent geometry. Reference response norms are not energy fractions; incoming covariance and omitted metric fluctuations remain explicit. The boson owner retains the finite projections, state assumptions and records.

A.5. Dense history

The retained DOP853 dense-polynomial history in `771c5a78` supplies source-compatible metric/field samples for probe evolution without replacing the nonlinear solution by sparse interpolation or extrapolation. The history record keeps its producer and regeneration route. This computational representation does not add a new source history or a physical epoch-selection rule.

A.6. Schur memory

The retarded finite-sector reduction in `61124309` derives the memory kernel from the same canonical block evolution. Its noise requires an incoming covariance, and initial cross-correlations remain physical data. A memory kernel alone is not a vacuum stress. The owner preserves that distinction and the original numbers.

A.7. Kick and shear

The York-constrained momentum preparations and amplitude-independent shear/internal-coupling relation in `b69dda73` remain finite conditional results. Their source profiles, zero freely supplied transverse-traceless datum and vacuum-reference curvature are explicit assumptions. The kick owner preserves the full derivation and tests.

A.8. Epsilon=.25 controls

The three-resolution control in `2ed1aea0` preserves the original nonspherical preparation and fixed proper endpoint, with decreasing constraint and adjacent readout differences. Its spreads are empirical indicators, not certified physical errors. The saved comparison is retained unchanged. The separate $\varepsilon=1$ nonlinear runs are **under-resolved, not used for claims**; they establish no trapped threshold.

A.9. Matter-work ledgers

The perturbative balance in `6561d8cf`, passive nonlinear ledger in `e6f34eb7`, and finite controls in `2ed1aea0` retain pressure, geometry, measure and boundary contributions rather than identifying energy loss with vacuum production. Their residuals remain numerical controls in the coupled-response and GH owners. The full `v0.26.0` technical atlas and all historical chapters are preserved in the archived foundation volume and in paper/technical-atlas-record.md.